

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

Name: Akanksha Kumari

Roll No: 2501010076

Email: 2501010076@krmu.edu.in

Q1. 1.4 Algorithm Execution Observation Table (5 marks)

```
import math
def generate_execution_observation_table(sizes):
    def fib_calls(n):
        if n<=1:
            return 1
        a,b=1,1
        for _ in range(2,n+1):
            a,b=b,1+a+b
        return b
    table=[
        "Algorithm Execution Observation Table",
        "InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci IterativeFibonacci
LinearSearch BinarySearch BubbleSort InsertionSort"
    ]
    for n in sizes:
        n=int(n)
        rec_fact=n+1
        iter_fact=n
        rec_fib=fib_calls(n)
        iter_fib=n
        lin_search=n
        bin_search=math.floor(math.log2(n))+1
        bubble=n*(n-1)//2
        insertion=n*(n-1)//2
        row=f"{n} {rec_fact} {iter_fact} {rec_fib} {iter_fib} {lin_search} {bin_search}
{bubble} {insertion}"
        table.append(row)
    # for line in table:
    #     print(line)
    return table
```